

The cycle type of two rows of a random Latin square: an exact ladder identity and a conditional proof of the weak Cavenagh–Greenhill–Wanless conjecture

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Abstract

Let L be a uniformly random Latin square of order n and σ the permutation of columns carrying its first row to its second. Cavenagh, Greenhill and Wanless conjectured that the cycle type of σ is asymptotically that of a uniform derangement, in total variation. We prove three exact identities — the marked-pair identity, which expresses the ratio of completion counts of adjacent cycle types as the odds of one bit (“rows 1, 2 lie in a common column cycle of a marked column pair”) in a single uniform square; a reduction theorem, by which the conjecture follows from that bit being fair to $o(1/\log n)$; and the ladder identity, by which the bit is exactly fair except on the squares in which no “ladder pair” of rows is flippable — and an orbit method by which the probability of the exceptional set is bounded by a product of conditionally independent factors. The conjecture is thereby reduced to a genericity hypothesis about the cycle structure of two rows at four columns (Hypothesis 5.4), which we state precisely, whose ingredients we test numerically up to $n = 100$, and which we locate within the known switching technology. Along the way we observe that the published proof of the upper bound in Lemma 3.12 of Cavenagh–Greenhill–Wanless has a gap for every split other than $(2, 2)$; our results do not depend on that bound, and we re-derive from the surviving direction the tail estimate on the number of cycles that the reduction theorem needs.

1 Introduction

Let P_n be the law of the cycle type of $\sigma = \sigma_{1,2}(L)$ for a uniform Latin square L of order n , and Q_n the law of the cycle type of a uniform derangement of $[n]$. Cavenagh, Greenhill and Wanless [1, Conj. 6.1] conjectured:

Conjecture 1.1. $d_{\text{TV}}(P_n, Q_n) = o(1)$ as $n \rightarrow \infty$.

For $\lambda \vdash n$ with all parts ≥ 2 let $C_n(\lambda)$ be the number of completions of a fixed $2 \times n$ Latin rectangle of cycle type λ to a Latin square, divided by $(n-2)!$ (the count does not depend on the rectangle chosen, since rectangles of the same type are isotopic by a row-preserving isotopy), and let $\gamma(\lambda)$ be the number of permutations of type λ . Then $P_n(\lambda) \propto \gamma(\lambda)C_n(\lambda)$ and $Q_n(\lambda) \propto \gamma(\lambda)$, so the conjecture says that C_n is nearly constant on the types that carry most of the mass. CGW proved $C_n(\lambda)/C_n(\mu) \in [\frac{1}{2}, \frac{3}{2}]$ for adjacent types (one part of λ split into two parts of μ , both ≥ 2), and used it to show that the two laws agree within a factor $n^{3/2}$ on every type with at most $\frac{6}{5} \log n$ parts; the upper bound $\frac{3}{2}$, and with it the upper half of the latter statement, is affected by the gap described in Remark 3.4 and [2].

Results. Throughout, a *mark* is a column pair $P = \{j, j'\}$ with $j' = \sigma^\alpha(j)$ on an m -cycle of σ , $m = \alpha + \beta$, $\alpha, \beta \geq 2$; $X = X(n, \lambda; \alpha, \beta)$ is the set of marked squares of type λ ; the *frame* of P is the permutation of rows $\pi(r) = r'$ with $L(r', j') = L(r, j)$; and P is a *B-pair* if rows 1, 2 lie in one cycle of π , an *A-pair* otherwise. Our results are:

- (1) *Marked-pair identity* (Theorem 2.4): $2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ exactly, for every $n, \lambda, \alpha, \beta$.
- (2) *Reduction* (Theorem 3.3): if $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \delta(n)/4$ for all adjacent pairs with $\alpha \neq \beta$, then $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. The tail estimate it needs on the number of cycles of σ is proved from the trivial half of (1) alone (Proposition 3.2).
- (3) *Ladder identity* (Theorem 4.2): with the ladder pairs $(x_k, y_k) = (\pi^{-k}(1), \pi^{-k}(2))$, $1 \leq k < K_0$, and “flippable” meaning that j, j' lie in different cycles of the column permutation of rows x_k, y_k ,

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad |\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{2} \mathbb{P}_X[\text{Flip} = \emptyset].$$

So Conjecture 1.1 follows from (L): $\mathbb{P}_X[\text{Flip} = \emptyset] = o(1/\log n)$.

- (4) *Orbit method* (Proposition 5.3): the flippabilities of a family of ladder pairs are conditionally independent given the orbit of a group of commuting column-cycle turns, and $\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X \exp(-\frac{1}{2} \sum_{k \in I} \bar{y}_k)$, where $\bar{y}_k$ is an explicit “orbit weight”. Hence (L), and the conjecture, follow from Hypothesis 5.4 (Proposition 5.5).

Hypothesis 5.4 is a statement of the form “two pairs of lines of a random Latin square are interleaved in the cycle structure of a third pair with probability bounded away from 0 and 1”, under conditioning on the cycle type of rows 1, 2. Without that conditioning its upper half is elementary and its lower half reduces to the row pair having macroscopic cycles (§6); the data give the interleaving probability 0.19 in every version we measured, and $\mathbb{P}_X[B] = 0.49\text{--}0.50$ (± 0.01) at $n = 30, 50, 100$. Section 6 explains why the switching methods available inside X — all of which we have pushed as far as they go — produce only fair identities and cannot by themselves give the hypothesis.

A gap in CGW’s Lemma 3.12. Case 3 of the splitting procedure in [1] (cross-switch at $\{\omega j, \omega j'\}$, then backflip at $\{j, j'\}$) does not behave as claimed: the cross-switch reverses the ω -arc from ωj to $\omega j'$ (or the complementary arc, depending on the smallest corner symbol), which contains j' (resp. j), so $\{j, j'\}$ ends at distance 2. For $\min(\alpha, \beta) \geq 3$ the backflip then lands in the wrong type; for $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right but the edge is not an edge of the joining multigraph, so the identity $G_S = G_J$ on which the double count rests fails (already at $n = 5$, where we computed both multigraphs exactly). We give the details in a separate note [2]; here we only record (Remark 3.4) which published statements depend on it. Nothing in the present paper does: the lower bound $\frac{1}{2}$ of CGW’s lemma is the trivial half of Theorem 2.4, and the upper bound $\frac{3}{2}$ is, by the same theorem, $\mathbb{P}_X[B] \leq \frac{2}{3}$ — a weak form of what we are after. By Theorem 4.2 it would follow from $\mathbb{P}_X[\text{Flip} = \emptyset] \leq \frac{1}{3}$.

Conventions. $L(r, c)$ is the entry in row r , column c . For rows x, y , $\rho_{x,y}$ is the permutation of columns with $L(y, \rho_{x,y}(q)) = L(x, q)$, so that $\sigma = \rho_{1,2}$ up to inversion; a *row cycle* of (x, y) is a cycle of $\rho_{x,y}$. For columns q, c , $\delta_{q,c}$ is the permutation of rows with $L(\delta(r), c) = L(r, q)$,

and its cycles are the *column cycles* of (q, c) . A *trade* of rows x, y along a row cycle swaps their entries on the columns of that cycle; a *turn* of a column cycle swaps the entries of columns q, c in its rows. Both are Latin trades (the result is again a Latin square). For $\lambda \vdash n$, $S(\lambda)$ is the set of Latin squares of order n with first row the identity whose rows 1, 2 have cycle type λ , so $|S(\lambda)| = \gamma(\lambda)(n-2)!C_n(\lambda)$; we also write $D_\lambda = \gamma(\lambda)$ for the number of derangements of type λ . A column pair $\{j, j'\}$ is *admissible* if $j' \notin \{j, \sigma(j), \sigma^{-1}(j)\}$ (CGW's condition for the pair operations; every marked pair is admissible since $\alpha, \beta \geq 2$). Following CGW, the *flip* at an A -pair whose columns lie in different σ -cycles, and the *backflip* (or *unflip*) at an A -pair whose columns lie in one σ -cycle, is the turn of the column cycle of the pair through row 2 (it joins, resp. splits, the σ -cycles involved); the *switch* at a pair whose columns lie in different σ -cycles is the trade of rows 1, 2 along the row cycle through one of the two columns, selected by CGW's rule (the cycle whose columns carry the smaller minimum symbol). We call Conjecture 1.1, which asks only for convergence in total variation, the *weak* conjecture; the stronger statements suggested by Cameron ("tends very rapidly") and by the data (pointwise ratios $1 + o(1)$) are not addressed here. Jacobson–Matthews sampling [3] is used for all numerical data; the scripts and logs are in the project repository.

2 The marked-pair identity

Throughout, squares are normalised to have first row id; CGW's column map ω , defined by $L(1, \omega(j)) = L(2, j)$, is then $\omega(j) = L(2, j)$, and we take $\sigma = \omega$ (only the cycle type matters, and it is the same for ω and ω^{-1}). For the proofs it is convenient to work with *unnormalised* squares (equivalently, symbol-relabelling classes): every instance count below then acquires a common factor $n!$, which cancels from every ratio, and the maps we use (flip, switch) act on unnormalised squares without any renormalisation step. Statuses, cycle types, and column positions are relabelling-invariant, so we freely state results in the normalised picture.

Definition 2.1 (marked instances). Let $\lambda \vdash n$ contain a part $m = \alpha + \beta$ ($\alpha, \beta \geq 2$), and let μ be λ with one part m split into α, β . The *splitting instance space* is

$$X = X(n, \lambda; \alpha, \beta) = \{(L', P) : L' \in S(\lambda), P = \{j, \sigma^\alpha(j)\} \text{ with } j \text{ in an } m\text{-cycle of } \sigma(L')\},$$

unordered pairs counted once. Thus every L' carries exactly $\lambda_m m$ marked pairs when $\alpha \neq \beta$ and $\lambda_m \alpha$ when $\alpha = \beta$. Since $\alpha, \beta \geq 2$ every marked pair is admissible, and $X = X_A \sqcup X_B$ according to the A/B status of P in L' . The *joining instance space* is

$$J = J(n, \mu; \alpha, \beta) = \{(L, P) : L \in S(\mu), P = \{j, j'\}, j \text{ in an } \alpha\text{-cycle}, j' \text{ in a different } \beta\text{-cycle}\},$$

again with $J = J_A \sqcup J_B$ by the status of P in L .

The counting arithmetic is immediate: with D_ν the number of derangements of cycle type ν and $N(\nu) = C_n(\nu)(n-2)!$,

$$\frac{2|X|}{|J|} = \frac{2 D_\lambda N(\lambda) \lambda_m m}{D_\mu N(\mu) \mu_\alpha \mu_\beta \alpha \beta} = \frac{2 C_n(\lambda)}{C_n(\mu)} \quad (\alpha \neq \beta), \quad (1)$$

using $D_\lambda/D_\mu = \alpha \mu_\alpha \beta \mu_\beta / (m \lambda_m)$; for $\alpha = \beta$ one has $D_\lambda/D_\mu = \alpha^2 \mu_\alpha (\mu_\alpha - 1) / (m \lambda_m)$, $|J| = |S(\mu)| \binom{\mu_\alpha}{2} \alpha^2$ and $|X| = |S(\lambda)| \lambda_m \alpha$, and (1) holds verbatim. The content of the identity is therefore the following count.

Lemma 2.2 (cycle trades preserve pair status). *Let $P = \{j, j'\}$ be an admissible column pair of a Latin square L and let C be one column cycle of P . Trading C (swapping the entries of columns j, j' in the rows of C) inverts the pair permutation π_P on C and fixes it elsewhere; the partition of rows into column cycles of P is unchanged. In particular the A/B status of P itself is invariant under the CGW flip at P , which trades the full column cycle through row 2.*

Proof. Write $\pi = \pi_P$, so $\pi(r)$ is the row of column j' holding the symbol $L(r, j)$; C is π -closed and the trade swaps $L(r, j) \leftrightarrow L(r, j')$ for $r \in C$. Fix $r \in C$ and let π' be the new pair permutation. The new $L(r, j)$ is the old $L(r, j')$; the row of the new column j' holding that symbol is found by solving (new $L(s, j')$) = (old $L(r, j')$): for $s \in C$ the new $L(s, j')$ is the old $L(s, j)$, and old $L(s, j) = \text{old } L(r, j')$ exactly when $\pi(s) = r$. Hence $\pi' = \pi^{-1}$ on C ; off C nothing changes. A cycle inverted in place has the same support, so the partition into cycles is unchanged. $\square$

Lemma 2.3 (switch toggle). *Let $(L, P) \in J$ and let Q be the row cycle through one column of P (the two row cycles are distinct), selected by CGW's min-symbol rule. Trading Q (the CGW switch) changes π_P by precomposition with the transposition of rows 1, 2: the cycles of π_P through rows 1, 2 are split if rows 1, 2 are in one cycle, and joined if in two. Hence the switch toggles the status of P , preserves $S(\mu)$ and the instance data (it inverts one σ -cycle in place), and is an involution: the trade preserves the traded cycle's symbol set (the two rows carry the same symbols on its columns), so the min-symbol rule re-selects the same cycle. Consequently $|J_A| = |J_B|$, exactly.*

Proof. Say Q is the row cycle through j ; its columns include j but not j' . The trade swaps the row-1 and row-2 entries in the columns of Q ; for π_P only column j matters, where the entries of rows 1, 2 are exchanged. Thus $\pi'_P = \pi_P \circ \tau_{12}$, and multiplying by a transposition splits or joins the cycles through its two points. Rows 1, 2 lie in one cycle of π_P exactly when P is a B -pair. The trade swaps the two rows' entries along the columns of Q , which inverts the corresponding σ -cycle in place: the cycle type and the column sets of all row cycles are preserved, so $(L, P) \mapsto (\text{sw}(L), P)$ maps J to J . $\square$

Theorem 2.4 (marked-pair identity). *For every n , λ , and split $m = \alpha + \beta$ with $\alpha, \beta \geq 2$ (including $\alpha = \beta$),*

$$|J_A| = |J_B| = |X_A|, \quad \text{hence} \quad \boxed{\frac{C_n(\lambda)}{C_n(\mu)} = \frac{|X|}{2|X_A|}, \quad 2 \frac{C_n(\lambda)}{C_n(\mu)} - 1 = \frac{|X_B|}{|X_A|}.}$$

In particular $C_n(\lambda) > \frac{1}{2} C_n(\mu)$ unconditionally, and Hypothesis 3.1 (EQ(δ)) is equivalent to

$$\left| \mathbb{P}(P \text{ is a } B\text{-pair}) - \frac{1}{2} \right| \leq \frac{\delta}{4}(1 + o(1)),$$

for (L', P) uniform on X : the law of a single bit — whether rows 1, 2 lie in one column cycle of the marked pair $\{j, \sigma^\alpha(j)\}$ — of a single uniform square of $S(\lambda)$.

Proof. By (1) it suffices to prove $|J| = 2|X_A|$. The flip at an A -pair P whose columns lie in one row cycle at σ -distances (α, β) splits that cycle into an α - and a β -cycle and maps $S(\lambda) \rightarrow S(\mu)$; by Lemma 2.2 the pair P is still an A -pair afterwards, so $x \mapsto \text{unflip}(x)$ maps X_A into J_A . The flip at a fixed pair is an involution (the traded cycle again contains row 2, with the same row support), so $\text{join} = \text{flip}$ inverts it: $X_A \leftrightarrow J_A$ is a bijection. Composing with the switch of Lemma 2.3 gives a bijection $X_A \leftrightarrow J_B$, namely $x \mapsto \text{sw}(\text{unflip}(x))$ with inverse $(L, P) \mapsto \text{flip}(\text{sw}(L))$ — the map “switch-then-flip”, which is exactly the CGW joining step at a B -pair. Hence $|J| = |J_A| + |J_B| = 2|X_A|$. $\square$

3 The reduction theorem

Hypothesis 3.1 (B-pair equidistribution, EQ(δ)). *For every $\lambda \vdash n$ with a part $m = \alpha + \beta$, $\alpha \neq \beta$, $\alpha, \beta \geq 2$, and μ the split type, $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq \delta(n)$, where $\bar{q}(\mu \rightarrow \lambda) := 2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ by Theorem 2.4.*

By Theorem 2.4, EQ(δ) is the statement $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{\delta}{4}(1 + o(1))$ for all adjacent pairs with $\alpha \neq \beta$. The proof below needs a tail bound on the number κ of cycles of σ ; we first prove it from the trivial half of Theorem 2.4 alone, so that nothing depends on the upper bound of [1, Lemma 3.12] (Remark 3.4).

Proposition 3.2 (the number of cycles, from the surviving direction only). *Let L be a uniform Latin square of order n , σ the column permutation of rows 1, 2, N_ℓ its number of ℓ -cycles and $\kappa = \sum_\ell N_\ell$. Then, using only the identity $|J| = 2|X_A|$ of Theorem 2.4 and $|X_A| \leq |X|$:*

(i) $\mathbb{E}[N_\ell] \leq 2n/(n - \ell)$ for every $\ell < n$; in particular $\mathbb{E}[N_\ell] \leq 4/\ell$ for $\ell \leq n/2$.

(ii) For every $k \geq 1$, with $(\kappa)_k = \kappa(\kappa - 1) \cdots (\kappa - k + 1)$,

$$\mathbb{E}[(\kappa)_k] \leq 2^{k-1} \sum_{\ell_1, \dots, \ell_k \geq 2} \frac{\mathbb{E}[mN_m]}{\ell_1 \cdots \ell_k} \Big|_{m=\ell_1+\dots+\ell_k} \leq 2^{k-1} \left(4 + \frac{k^2}{\log n}\right) (\log n)^k.$$

(iii) For $A \geq 3$, $\mathbb{P}[\kappa \geq A \log n] \leq (7 + \log n) n^{-\log((A-1)/2)}$, which is $O(n^{-2} \log n)$ for $A \geq 16$; and $\mathbb{P}[\text{all cycles of } \sigma \text{ have the same length } \ell < n] \leq 4d(n)/n$, $d(n)$ the number of divisors.

Proof. (i) For $\beta \neq \ell$ the joining instances $(L, \{j, j'\})$ with j in an ℓ -cycle and j' in a β -cycle number $\ell N_\ell \beta N_\beta$ per square, the splitting instances (pairs at distance ℓ on an $(\ell + \beta)$ -cycle) number $(\ell + \beta) N_{\ell+\beta}$, and $|J| = 2|X_A| \leq 2|X|$ gives $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta) N_{\ell+\beta}]$; for $\beta = \ell$ the same count with $\ell^2 \binom{N_\ell}{2}$ and $\ell N_{2\ell}$ gives $\mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 4\ell \mathbb{E}[N_{2\ell}]$. Summing the first over $\beta \neq \ell$ ($\sum_\beta \beta N_\beta = n$) and adding the second (its right side is the missing summand $m = 2\ell$): $\mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 2\mathbb{E}[\sum_{m \geq \ell+2} m N_m] \leq 2n$. Hence $n\mathbb{E}[\ell N_\ell] = \mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] + \ell \mathbb{E}[\ell N_\ell] \leq 2n + \ell \mathbb{E}[\ell N_\ell]$.

(ii) Count the instances $(L, (C_1, j_1), \dots, (C_k, j_k))$: C_i pairwise distinct σ -cycles of L of lengths ℓ_i and $j_i \in C_i$; there are $\ell_1 \cdots \ell_k$ of them per ordered k -tuple of distinct cycles with these lengths. Join C_1 and C_2 at the pair $\{j_1, j_2\}$ by a joining step: flip if the pair is an A -pair; if it is a B -pair, first trade the row cycle C_2 of j_2 (a switch in the sense of Lemma 2.3, whose proof applies to the row cycle through either column of the pair; it inverts C_2 in place and does not touch the cycle of j_1), then flip. The flip replaces σ by $\sigma \circ (j_1 j_2)$ on rows 1, 2, so the result has j_1, j_2 on one cycle of length $\ell_1 + \ell_2$ with $j_2 = \sigma^{\ell_2}(j_1)$ whether or not the trade was applied (CGW Lemma 3.3), every other cycle of σ (in particular $C_3, \dots, C_k$) untouched, and the map is at most two-to-one (the backflip at $\{j_1, j_2\}$ returns L or the traded square). Repeat with $\{j_1, j_3\}, \dots, \{j_1, j_k\}$, always trading the cycle of the new column when needed, so that the cycle containing j_1 is never inverted: after step i every earlier j_r is shifted by exactly ℓ_{i+1} . After $k - 1$ steps all j_i lie on one cycle of length $m = \sum \ell_i$ at the positions $j_i = \sigma^{\ell_i + \dots + \ell_k}(j_1)$, determined by j_1 and $(\ell_1, \dots, \ell_k)$, and the composite is at most 2^{k-1} -to-one. (With CGW's min-symbol rule the trade may invert the merged cycle and the positions are not determined; the count would then be 2^{2k-3} -to-one.) Hence the number of instances with lengths (ℓ_i) is at most $2^{k-1} \sum_{L'} m N_m(L')$, i.e. $\mathbb{E}[\ell_1 \cdots \ell_k T(\ell_1, \dots, \ell_k)] \leq 2^{k-1} \mathbb{E}[m N_m]$ with T the number of

ordered k -tuples of distinct cycles with these lengths; summing over (ℓ_i) gives the first bound, since $\sum_{(\ell_i)} T = (\kappa)_k$. For the second, split at $m \leq n/2$, where $\mathbb{E}[mN_m] \leq 4$ by (i) and $\sum_{m \leq n/2} \sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq (\sum_{\ell \leq n/2} \ell^{-1})^k \leq (\log n)^k$, and $m > n/2$, where some $\ell_i \geq m/k$, so $\sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq k \cdot (k/m)(\log n)^{k-1}$, and $\sum_{m > n/2} mN_m \cdot k^2(\log n)^{k-1}/m \leq k^2(\log n)^{k-1}$ because at most one cycle is longer than $n/2$.

(iii) Markov for $(\kappa)_k$ with $k = \lceil \log n \rceil \leq \log n + 1$: on $\{\kappa \geq K\}$, $(\kappa)_k \geq (K - k + 1)^k \geq ((A - 1) \log n)^k$ for $K = A \log n$, and $k^2 / \log n \leq \log n + 3$, so $\mathbb{P}[\kappa \geq K] \leq 2^{k-1}(4 + k^2 / \log n)(\log n)^k / ((A - 1) \log n)^k \leq (7 + \log n)(2/(A - 1))^k \leq (7 + \log n)(2/(A - 1))^{\log n}$ for $A \geq 3$, and $(A - 1)/2 \geq e^2$ for $A \geq 16$. For the equal-length types, $\ell \mid n$ and $\ell < n$ force $\ell \leq n/2$, so $\mathbb{P}[N_\ell = n/\ell] \leq \mathbb{E}[N_\ell] \ell / n \leq 4/n$; $\ell = n$ is the n -cycle. $\square$

Theorem 3.3. Assume EQ(δ) with $\delta = \delta(n) \rightarrow 0$. Then

$$d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)}).$$

In particular EQ($o(1/\log n)$) implies Conjecture 1.1.

Proof. Call λ *plain* if not all its parts are equal (so the only non-plain types are (ℓ^κ) , $\kappa \geq 2$).

Routing. Every plain λ with $\kappa = \kappa(\lambda)$ parts is reachable from (n) by a chain of $\kappa - 1$ splits, each with $\alpha \neq \beta$. Build the chain in reverse, splitting parts of λ off a shrinking residual, in a suitable order; a collision means splitting a part p off a residual R with $p = R - p$. If the two largest parts of λ are unequal, split smallest-first and let them be the final pair: while at least three parts remain, the split-off minimum p satisfies $R - p \geq 2p > p$ (all parts are ≥ 2 and at least two other parts remain), so no collision occurs, and the final split is unequal by hypothesis. If instead the two largest parts are equal, say to $\ell_{\max}$, then $\ell_{\max} \neq n/2$ (else $\lambda = (n/2, n/2)$ is not plain), so we may split one copy of $\ell_{\max}$ off first without collision, then proceed smallest-first among the remaining parts, reserving for the final split the pair $(b, \ell_{\max})$ where $b < \ell_{\max}$ is the smallest part: at every intermediate step the residual minus the split-off minimum still contains the reserved $\ell_{\max}$ and the reserved b , hence exceeds the minimum (also when a further copy of $\ell_{\max}$ is the split-off minimum: $R - p \geq \ell_{\max} + b > p$), and the final split is unequal.

For each step of such a chain, Theorem 2.4 and EQ(δ) give $|S(p_{i+1})|/|S(p_i)| = (\gamma(p_{i+1})/\gamma(p_i))(1 + \varepsilon_i)$ with $|\varepsilon_i| \leq \delta$. Multiplying along the chain,

$$\frac{P_n(\lambda)}{P_n((n))} = \frac{\gamma(\lambda)}{\gamma((n))} \prod_{i=1}^{\kappa-1} (1 + \varepsilon_i)^{\pm 1}, \quad |\varepsilon_i| \leq \delta,$$

so for all plain λ with $\kappa(\lambda) \leq K$, $P_n(\lambda) = Q_n(\lambda) Z^{-1}(1 + O(K\delta))$ (the bound is vacuous unless $K\delta = O(1)$, which is the case of interest) where $Z > 0$ is a normalising constant independent of λ (namely $Z = \sum_{\nu} \gamma(\nu) C_n(\nu) / (|\mathcal{D}_n| C_n((n)))$).

Let $K = \lceil A \log n \rceil$ with A a large absolute constant, and let $\mathcal{B}$ be the set of types that are either non-plain or have $\kappa > K$. Under Q_n , Markov's inequality with the generating function $\sum_{\pi \in \mathcal{D}_n} 2^{\kappa(\pi)} = n! [x^n] e^{-2x} (1 - x)^{-2} \leq n \cdot n!$ (cf. [1, proof of Lemma 4.4]) gives $Q_n(\kappa > K) \leq e n 2^{-K} = O(n^{-2})$ for $A \log 2 \geq 3$; also $Q_n(\text{non-plain}) \leq \sum_{\ell \mid n, \ell < n} \gamma((\ell^{n/\ell})) / D_n = O(n^{-1})$. Under P_n , Proposition 3.2 gives $P_n(\kappa > K) = O(n^{-2} \log n)$ for $A \geq 16$ and $P_n(\text{non-plain}) = O(n^{-1+o(1)})$ without any input from [1] beyond the identity of Theorem 2.4.

On the complement of $\mathcal{B}$, summing $P_n = Q_n Z^{-1}(1 + O(K\delta))$ over $\lambda \notin \mathcal{B}$ and using the tail bounds shows $Z^{-1} = 1 + O(K\delta) + O(n^{-1+o(1)})$, whence $\sum_{\lambda \notin \mathcal{B}} |P_n - Q_n| = O(K\delta) + O(n^{-1+o(1)})$. Combining with the tails, $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. $\square$

Remark 3.4 (the gap in CGW’s Lemma 3.12). In [1], the direction $|S(\lambda, F)|N_\lambda \geq \frac{1}{2}|S(\mu, F)|N_\mu$ of Lemma 3.12 uses only that each splitting pair carries at most two edges and each joining pair at least one; by Theorem 2.4 it is the trivial inequality $|X| \geq |X_A|$. The direction $\leq \frac{3}{2}$ requires every splitting pair of every λ -square to carry exactly two edges that are also edges of the joining multigraph, and case 3 of the splitting procedure fails to supply them for every split other than $(2, 2)$ [2]. Consequently the upper bound of CGW’s Theorem 3.13, the lower bound of their Lemma 4.1, the parity bounds of Theorem 4.3, the bound $\mathbb{P}[\sigma \text{ is an } n\text{-cycle}] \leq 2n^{-2/3}$ of Lemma 4.4, both halves of Corollary 4.5, and the results of their Section 4.2 that rest on these (Corollary 4.6, Lemma 4.7, Theorems 4.9 and 4.11, Corollary 4.10 — the last superseded by [4]) are not established by the published argument. The present paper uses none of them: Proposition 3.2 replaces Corollary 4.5 in the proof of Theorem 3.3. The exact completion counts tabulated in [1] give $C_n(\lambda)/C_n(\mu) \in [0.985, 1.0035]$ for every adjacent pair at $7 \leq n \leq 11$ (and $= \frac{3}{2}$ at $n = 5$, $\frac{7}{8}$, $\frac{11}{14}$, $\frac{21}{22}$ at $n = 6$), so the statement itself is not in doubt.

4 The ladder identity

Notation: $(L, P) \in X = X(n, \lambda; \alpha, \beta)$, $P = \{j, j'\}$, frame $\pi = \pi_P$ ($L(\pi(r), j') = L(r, j)$), C_r the π -cycle of row r , $\rho_{x,y}$ the column permutation of rows x, y ($L(y, \rho_{x,y}(q)) = L(x, q)$), (x, y) *flippable* iff j, j' lie in different cycles of $\rho_{x,y}$, and the j' -trade at (x, y) exchanges rows x, y on the $\rho_{x,y}$ -cycle through j' , which replaces π by $(xy) \circ \pi$ if (x, y) is flippable and by $(xy) \circ \pi \circ (xy)$ otherwise. $B = \{1 \sim 2\}$ (rows 1, 2 in a common frame cycle), A its complement, $\mathbf{1}_B$ the bit; for B -instances d_{12} and d_{21} are the numbers of π -steps from 1 to 2 and from 2 to 1.

Definition 4.1 (ladder). Put $K_0 = \min(|C_1|, |C_2|)$ on A and $K_0 = \min(d_{12}, d_{21})$ on B , and for $1 \leq k < K_0$ let

$$x_k = \pi^{-k}(1), \quad y_k = \pi^{-k}(2), \quad T_k = \text{the } j'\text{-trade at } (x_k, y_k).$$

The pairs $\{x_k, y_k\}$ are the *ladder* of (L, P) , and $\text{Flip}(L, P) = \{k < K_0 : (x_k, y_k) \text{ flippable}\}$.

The rows $x_1, \dots, x_{K_0-1}$ are distinct rows of $C_1 \setminus \{1\}$ (on A) or of the arc from 2 to 1 (on B), the y_k likewise in $C_2 \setminus \{2\}$ or the arc from 1 to 2; so the $2(K_0 - 1)$ ladder rows are distinct and avoid rows 1, 2, and $K_0 \geq 2$: on A because π is a derangement, on B by the admissibility of the mark ($\pi(1) \neq 2$, $\pi(2) \neq 1$).

Theorem 4.2 (ladder identity). *For every n , λ and split $\alpha + \beta = m$ with $\alpha, \beta \geq 2$,*

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad \text{hence} \quad |\mathbb{P}_X[B] - \tfrac{1}{2}| \leq \tfrac{1}{2}\mathbb{P}_X[\text{Flip} = \emptyset].$$

The same identity holds with X replaced by the set of squares with a fixed pair of rows 1, 2 (of type λ) and a fixed mark P .

Proof. Fix $(L, P) \in X$ and write $K = \{1, \dots, K_0 - 1\}$. The ladder depends on (L, P) , so the T_k are not maps of X ; the statement to prove is that $(L, P) \sim T_S(L, P)$, $S \subseteq K$ (the trades at the unordered pairs of S , in any order), is an equivalence relation on X whose classes have exactly $2^{|K|}$ elements and on which the bit is $\mathbf{1}_B(T_S(L, P)) = \mathbf{1}_B(L, P) + |S \cap \text{Flip}| \pmod{2}$. This follows once we know that each T_k is an involution of X which preserves K_0 , the set of unordered ladder pairs and the flippability of each of them, that T_k toggles the bit exactly when $k \in \text{Flip}$, and that trades at disjoint row pairs commute: then the same trades, at the same unordered pairs with

the same flippabilities, are available from every point of a class (symmetry and transitivity), the $2^{|K|}$ points $T_S(L, P)$ are distinct (rows x_k, y_k are changed by T_k and by no other T_i), and the bit formula holds. If $\text{Flip} \neq \emptyset$ exactly half of the subsets S have $|S \cap \text{Flip}|$ odd, so B and A each contain half of the class; summing over classes gives the identity, and the bound follows from $\mathbb{P}[B] + \mathbb{P}[A] = 1$.

T_k is a well-defined involution of X . The j' -trade at (x_k, y_k) changes only rows $x_k, y_k \notin \{1, 2\}$, so σ and the mark are untouched and (L, P) stays in X ; it is an involution because the traded columns are again the ρ'_{x_k, y_k} -cycle through j' .

Effect on the frame. If $k \in \text{Flip}$ the new frame is $\pi' = (x_k y_k) \circ \pi$. On A this merges C_1 and C_2 : the new cycle runs $1 \rightarrow \pi(1) \rightarrow \dots \rightarrow x_{k+1} \rightarrow y_k \rightarrow \pi(y_k) \rightarrow \dots \rightarrow 2 \rightarrow \dots \rightarrow y_{k+1} \rightarrow x_k \rightarrow \dots \rightarrow 1$, so $d'_{12} = |C_1|$ and $d'_{21} = |C_2|$ (reading $x_{|C_1|} = 1, y_{|C_2|} = 2$). On B , x_k lies on the arc from 2 to 1 and y_k on the arc from 1 to 2 (as $k < \min(d_{12}, d_{21})$), and the split into $(x_k, \pi(x_k), \dots, \pi^{-1}(y_k))$ and $(y_k, \dots, \pi^{-1}(x_k))$ puts 1 (k steps after x_k , fewer than the d_{12} steps from x_k to y_k) in the first cycle and 2 in the second: the new state is in A with $|C'_1| = d_{12}$, $|C'_2| = d_{21}$. In both cases K_0 is preserved and the bit toggles. If $k \notin \text{Flip}$ the new frame is $(x_k y_k) \circ \pi \circ (x_k y_k)$: the cycle partition is unchanged up to exchanging the labels x_k, y_k , so K_0 and the bit are preserved.

The ladder is preserved. Let $\pi' = (x_k y_k) \circ \pi$ (the conjugation case is similar and easier). Since $\pi'^{-1}(r) = \pi^{-1}((x_k y_k)(r))$, the preimage chains of 1 and 2 under π' agree with those under π up to depth k and are exchanged beyond it: $\pi'^{-i}(1) = x_i$ and $\pi'^{-i}(2) = y_i$ for $i \leq k$, while $\pi'^{-(k+1)}(1) = \pi^{-1}(y_k) = y_{k+1}$ and then $\pi'^{-i}(1) = y_i, \pi'^{-i}(2) = x_i$ for $k < i < K_0$. So the unordered pairs $\{x_i, y_i\}$, $i \in K$, are the ladder of $T_k(L, P)$. In the conjugation case the same computation with $(x_k y_k) \circ \pi \circ (x_k y_k)$ gives $\pi'^{-i}(1) = (x_k y_k)(x_i)$, which again permutes the pairs $\{x_i, y_i\}$ among themselves (only $i = k$ is affected, by exchanging x_k, y_k).

Flippability is preserved and the T_k commute. T_k changes rows x_k, y_k only, and the flippability of (x_i, y_i) , $i \neq k$, depends only on rows x_i, y_i , which are different rows. Likewise T_i and T_k act on disjoint pairs of rows, each along a column set determined by its own two rows, so $T_i T_k = T_k T_i$. (After T_k the pair $\{x_i, y_i\}$ may have changed its index, but the trade at an unordered pair does not depend on the index or on the order of the two rows.)

The statement for fixed R holds because every T_k fixes rows 1 and 2. $\square$

Corollary 4.3 (the conjecture reduces to the ladder). *Write $f(n, \lambda; \alpha, \beta) = \mathbb{P}_X[\text{Flip} = \emptyset]$. By Theorem 2.4, $\bar{q}(\mu \rightarrow \lambda) - 1 = (\mathbb{P}_X[B] - \mathbb{P}_X[A])/\mathbb{P}_X[A]$ and $\mathbb{P}_X[A] \geq (1 - f)/2$, so $|\bar{q} - 1| \leq 2f/(1 - f)$: EQ(δ) holds with $\delta(n) = \max_{\lambda, \alpha \neq \beta} 2f/(1 - f)$. If this $\delta(n) \rightarrow 0$, Theorem 3.3 gives*

$$d_{\text{TV}}(P_n, Q_n) = O\left(\log n \cdot \max_{\lambda, \alpha \neq \beta} f(n, \lambda; \alpha, \beta) + n^{-2/3}\right).$$

In particular the weak CGW conjecture follows from

$$(L) \quad \mathbb{P}_X[\text{no ladder pair is flippable}] = o(1/\log n) \quad \text{uniformly in } \lambda, \alpha, \beta. \quad (2)$$

Remark 4.4 (data). On Jacobson–Matthews samples with a uniformly random mark (9000, 7200, 3000 ladder-pair instances at $n = 30, 50, 100$; runs/s13/f1/): $\mathbb{P}[\text{Flip} = \emptyset \mid K_0] = 2^{-(K_0-1)}$ within statistical error for $K_0 \leq 6$ and consistent with it beyond; $\mathbb{P}[\text{Flip} = \emptyset] = 0.125, 0.079, 0.041 \approx 4/n$; the ladder pairs are flippable with frequency 0.5002 (62082 pairs at $n = 50$); and $\mathbb{P}[B] - \mathbb{P}[A] = -0.022 \pm 0.011, +0.008 \pm 0.012, -0.030 \pm 0.018$.

4.1 Offset ladders and short arcs

The ladder is one of a family. For an integer c (an *offset*) put

$$K_c = \begin{cases} \min(|C_1| - c, |C_2|) & \text{on } A, \\ \min(d_{21} - c, d_{12}) & \text{on } B, \end{cases} \quad D_c = \{(x_{c+i}, y_i) = (\pi^{-(c+i)}(1), \pi^{-i}(2)) : 1 \leq i < K_c\},$$

so D_0 is the ladder and for $c > 0$ the x -row is c steps deeper than the y -row (x -depths $c + 1, \dots, c + K_c - 1$ stay inside $C_1 \setminus \{1\}$, resp. the arc from 2 to 1; y -depths $1, \dots, K_c - 1$ inside $C_2 \setminus \{2\}$, resp. the arc from 1 to 2). We take $c \geq 0$; the families with the y -row deeper are obtained by exchanging the roles of rows 1 and 2 (and of C_1, C_2, d_{12}, d_{21}) throughout.

Theorem 4.5 (offset ladders). *For every offset $c \geq 0$, on the domain $\mathcal{D}_c = \{K_c \geq 2\}$ (where $D_c \neq \emptyset$) the j' -trades at the pairs of D_c are commuting involutions that preserve K_c , the unordered pairs of D_c and their flippabilities, and toggle the bit iff the traded pair is flippable. Hence, with Flip_c the set of flippable pairs of D_c ,*

$$\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = \mathbb{P}[B, \mathcal{D}_c, \text{Flip}_c = \emptyset] - \mathbb{P}[A, \mathcal{D}_c, \text{Flip}_c = \emptyset].$$

Proof. Identical to Theorem 4.2; only the bookkeeping changes. Let $\pi' = (x_{c+i} y_i) \circ \pi$. The preimage chain of 1 under π' agrees with that of π up to depth $c+i$ and then continues along the chain of 2: $\pi'^{-(c+i+t)}(1) = y_{i+t}$; symmetrically $\pi'^{-(i+t)}(2) = x_{c+i+t}$ ($t \geq 1$). So the pair of D_c at y -depth $i' > i$ is $\{y_{i'}, x_{c+i'}\}$, unchanged as a set, and the pairs at depth $i' < i$ are untouched: D_c is preserved. (The same computation shows that trading a pair at depths (k, k') sends a deeper pair at depths (K, K') to depths $(K' + k - k', K - k + k')$, whose offset is $2(k - k') - (K - K')$; it is preserved iff $K - K' = k - k'$, so only families of constant offset, i.e. the D_c , are preserved.) On A the trade merges C_1, C_2 into a cycle with $d'_{12} = |C_1| - (c + i) + i = |C_1| - c$ and $d'_{21} = |C_2| + c$, so $K'_c = \min(d'_{21} - c, d'_{12}) = \min(|C_2|, |C_1| - c) = K_c$; on B the pair separates rows 1, 2 ($c + i < d_{21}$ puts x_{c+i} on the arc from 2 to 1, $i < d_{12}$ puts y_i on the arc from 1 to 2) and gives $|C'_1| = d_{12} + c$, $|C'_2| = d_{21} - c$, again $K'_c = \min(d_{12}, d_{21} - c) = K_c$. No pair of D_c is frame-adjacent: $y_i = \pi(x_{c+i})$ would need $d_{21} = c - 1$ (or $\pi(1) = 2$ when $c = 0$) and $x_{c+i} = \pi(y_i)$ would need $d_{21} = c + 1$, both excluded on $\mathcal{D}_c$. Commutation, preservation of flippability and the orbit count are as before. $\square$

Exact enumeration at $n = 7$ ($\lambda = (7)$, $\{\alpha, \beta\} = \{2, 5\}$) confirms the identities for $c = 0, 1, 2, 3$ to all digits ($\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = -0.005848, +0.001462, +0.003509, -0.001316$, equal to the right-hand sides).

Using the offsets to price short arcs. Fix $\ell \geq 2$ and consider B -instances with $d_{12} = \ell$. For $0 \leq c \leq n/2 - \ell$ and $d_{21} \geq c + \ell$ one has $K_c = \ell$, so D_c has $\ell - 1$ pairs, and the offset- c trade at a flippable pair of D_c lands in A with $|C_1| = \ell + c \leq n/2$ and $|C_2| = d_{21} - c \geq \ell$. Within a class of $\mathcal{D}_c$ all B -members have the same (d_{12}, d_{21}) and all A -members the same $(|C_1|, |C_2|) = (d_{12} + c, d_{21} - c)$, and Flip_c is a class invariant, so Theorem 4.5 class by class and Proposition 4.6 give

$$\mathbb{P}[B, d_{12} = \ell, d_{21} \geq c + \ell, \text{Flip}_c \neq \emptyset] = \mathbb{P}[A, |C_1| = \ell + c, |C_2| \geq \ell, \text{Flip}_c \neq \emptyset] \leq \frac{18}{n - \ell - c - 1} \leq \frac{36}{n - 2}. \quad (3)$$

Averaging over c : if $\theta(L, P)$ denotes the fraction of offsets $c \in [0, n/2 - \ell]$ whose diagonal D_c contains a flippable pair, then for every $\theta_0 > 0$

$$\mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0] \leq \frac{36}{(n-2)\theta_0}, \quad (4)$$

since on $|C_1| \geq n/2 + \ell$ one has $d_{21} = |C_1| - \ell \geq n/2 \geq c + \ell$ for every such c , so $\mathbb{E}[\theta \mathbf{1}_E]$ is the average over c of the left side of (3), and Markov's inequality applies. Together with the splice-in bound $\mathbb{P}[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$ for $\ell \leq n/4$ (Proposition 4.7 below), this prices every short arc except on the event $\{\theta < \theta_0\}$.

Proposition 4.6 (short frame cycles through a marked row are rare). *For (L, P) uniform on X and every $2 \leq \ell \leq n/2$,*

$$\mathbb{P}[|C_1| = \ell] \leq \frac{18}{n - \ell - 1}, \quad \text{hence} \quad \mathbb{P}[|C_1| \leq \ell] \leq \frac{36\ell}{n} \quad (\ell \leq n/2 - 1),$$

and the same bounds hold for C_2 .

Proof. Let $E_\ell = \{(L, P) \in X : |C_1| = \ell\}$. (The case $C_1 = \{1, 2\}$ cannot occur: it would mean $\sigma(j) = j'$ and $\sigma(j') = j$, but $j' = \sigma^\alpha(j)$ lies in an m -cycle with $m \geq 4$.)

Forward moves. From $(L, P) \in E_\ell$ choose $y \in C_1 \setminus \{1, 2\}$ and $x \notin C_1 \cup \{2\}$: at least $(\ell - 2)(n - \ell - 1)$ pairs $((\ell - 1)(n - \ell - 1)$ if $2 \notin C_1$; for $\ell = 2$ this is $n - 3$). If (x, y) is flippable, apply the j' -trade at (x, y) (trade rows x, y along the $\rho_{x,y}$ -cycle $Q_{x,y}$ through j'); otherwise turn C_1 (which makes (x, y) flippable, as $y \in C_1 \not\equiv x$) and then trade $Q_{x,y}$ in the turned square. In both cases the resulting frame is $(xy) \circ \pi^\pm$ with $\pi^\pm$ equal to π or to π inverted on C_1 , so the cycle of row 1 becomes $C_1 \cup C_x$, of length $\ell + |C_x| > \ell$; rows 1, 2 are untouched by the trade, and the result is in X (no turn, or $2 \in C_1$) or in J (turn with $2 \notin C_1$).

Backward moves. A square $(L', P) \in X \cup J$ is reached from $(L, P) \in E_\ell$ by the move at (x, y) only if x, y lie in the cycle C'_1 of row 1 in L' , and undoing the trade (the same trade, since $\rho'_{x,y} = \rho_{x,y}^{-1}$ on $Q_{x,y}$) splits C'_1 into the cycles $\{x, \pi'(x), \dots, \pi'^{-1}(y)\}$ and $\{y, \pi'(y), \dots, \pi'^{-1}(x)\}$ of $(xy) \circ \pi'$, of which the one containing row 1 must have length exactly ℓ : given L' , that arc is determined by its position relative to row 1 (ℓ choices), and the arc determines $\{x, y\}$. With the binary choice “turn back or not” this gives at most 2ℓ preimages per (L', P) .

Counting. The map $((L, P), \text{move}) \mapsto ((L', P), \text{inverse move})$ is injective, so $|E_\ell|(\ell - 2)(n - \ell - 1) \leq 2\ell(|X| + |J|) \leq 6\ell|X|$, using $|J| = 2|X_A| \leq 2|X|$ (Theorem 2.4). For $\ell \geq 3$ this is the claim; for $\ell = 2$ the forward count is $n - 3$ and the backward count 4, giving $12/(n - 3)$. The bound for C_2 is proved by the same argument with the roles of rows 1 and 2 exchanged: a turn of $C_2 \not\equiv 1$ replaces σ by $(\sigma(j)\sigma(j')) \circ \sigma$, which also splits the m -cycle into lengths α, β and lands in J . $\square$

Proposition 4.7 (the splice-in bound). *For $2 \leq \ell \leq n/4$, $\mathbb{P}_X[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$, and the same with d_{21} in place of d_{12} .*

Proof. Let E be the event. *Forward moves.* From $(L, P) \in E$ choose $y = \pi^i(1)$ with $1 \leq i \leq \ell - 1$ (a row strictly inside the arc from 1 to 2; $\ell - 1$ choices) and a row $x \notin C_1$ ($n - |C_1| \geq (n - 2\ell + 1)/2$ choices), and let C_x be the frame cycle of x , which contains neither 1 nor 2. If (x, y) is not flippable, turn C_x : this trades columns j, j' on the rows of C_x only, so it preserves X , inverts π on C_x (Lemma 2.2), fixes row y , and replaces $\rho_{x,y}$ by $\rho_{x,y} \circ (jj')$, which makes (x, y) flippable. Then apply the j' -trade at (x, y) , which replaces the frame by $(xy) \circ \pi^\pm$ ($\pi^\pm = \pi$, or π inverted

on C_x): the arc from 1 to 2 becomes $1 \rightarrow \dots \rightarrow \pi^{-1}(y) \rightarrow x \rightarrow \dots \rightarrow y \rightarrow \dots \rightarrow 2$ with C_x occupying positions $i, \dots, i+|C_x|-1$, so d_{12} grows by $|C_x|$, rows 1, 2 are untouched, and the result is again in X_B with the same mark. *Backward moves.* Given the target $(L', P) \in X_B$ and ℓ , a preimage is determined by the position $i \in \{1, \dots, \ell-1\}$ of x on the arc of L' from 1 to 2 (then $|C_x| = d'_{12} - \ell$ and $y = \pi'^{i+|C_x|}(1)$ are determined), together with the binary choice of whether C_x was turned; undoing is the same j' -trade (the pair stays flippable, $\rho'_{x,y} = \rho_{x,y}^{-1}$ on the traded cycle) followed by the turn. So each target has at most $2(\ell-1)$ preimages while each source has at least $(\ell-1)(n-2\ell+1)/2$ forward moves, whence $|E|(\ell-1)(n-2\ell+1)/2 \leq 2(\ell-1)|X|$. $\square$

5 The orbit method

Throughout this section $X = X(n, \lambda; \alpha, \beta)$, and “legal” means: preserves the type of rows 1, 2 and the mark, i.e. maps X to itself. The mark is now written $\{p, p'\}$ (so $p = j$, $p' = j'$ in the notation of §§2–4; the letters j, j' are freed for other columns), π is its frame, and a column pair (q, c) is *free* if $\{q, c\} \cap \{p, p'\} = \emptyset$. A pair of rows (x, y) is called *parallel* if it is flippable (p, p' in different cycles of $\rho_{x,y}$) and *crossed* otherwise; Q_p and $Q_{p'}$ denote the $\rho_{x,y}$ -cycles through p and p' . A free pair (q, c) is *separated* by $\{p, p'\}$ in $\rho_{x,y}$ if q and c lie on different arcs of a common cycle through p, p' (crossed case) or one in Q_p and the other in $Q_{p'} \neq Q_p$ (parallel case). The inputs $(\alpha), (\beta), (\gamma)$ of the hypothesis are named in §6.

(a) Transposition: which half of the conditioning is the obstacle. Let L be uniform on *all* Latin squares of order n (no condition on rows 1, 2 in this paragraph), let x, y carry four distinct symbols on $\{p, p'\}$, and condition on the two full columns p, p' (the frame π of (p, p')). In the transposed square $L^\top$ the columns p, p' are the distinguished rows, the frame π is their column permutation, and the event “ p, p' lie in different cycles of $\rho_{x,y}$ ” is the event “ $\{x, y\}$ is an A -pair for the distinguished rows”. Conditioning on the two full rows p, p' of $L^\top$ is conditioning on their type and on the position of the pair $\{x, y\}$ in their cycle structure, which is free by symbol and column relabelling. Hence, by Lemma 2.3 ($|J_A| = |J_B|$, which needs no condition on the cycle lengths) and CGW’s Theorem 3.13 (the inequality $\mathbb{P}_X[A] \geq \frac{1}{3}$, whose published proof has a gap for every split other than $(2, 2)$, Remark 3.4):

$$\mathbb{P}[p, p' \text{ in different cycles of } \rho_{x,y} \mid \text{columns } p, p'] = \frac{1}{2} \text{ if } x, y \text{ are apart in the frame,} \quad \geq \frac{1}{3} \text{ if together at distance } \leq 1 \quad (5)$$

So the statement with the *columns* prescribed is CGW’s theorem; it is the conditioning on rows 1, 2 — on their type and the mark, which is what defines X and cannot be removed — that is the obstacle. In the same transposed picture the ladder trades T_k of Theorem 4.2 at flippable pairs are CGW’s flips and backflips at the column pairs $\{x_k, y_k\}$ of $L^\top$, and Theorem 4.2 is exactly the legal half of CGW’s Lemma 3.12 for the distinguished rows p, p' : the other half — the switch and the cross-switch at $\{x_k, y_k\}$ — trades the columns p, p' of L along a frame cycle or frame arc through exactly one of rows 1, 2 (apart: $C_1 \ni 1$, $C_1 \not\ni 2$; together: the arc from x_k to y_k contains 1 but not 2, the other arc 2 but not 1), so it changes the type of rows 1, 2 and leaves X . The switch is, in the original orientation, the unflip of rows 1, 2 at (p, p') (which maps $X_A \rightarrow J_A$); it toggles the crossed/parallel status of *every* pair (x, y) with x on the side of row 1 and y on the side of row 2, and reverses the orientation of one side, so it carries the ladder (a diagonal of the side-by-side array of pairs) to a different family (an anti-diagonal). The cross-switch likewise maps X_B to J_B (it splits the σ -cycle through p, p' while keeping rows

1, 2 in one frame cycle, CGW Lemma 3.10) and toggles the interior pairs. Both are exact but not usable for the ladder bits in X .

(b) Row trades are free in X ; column turns are conditionally free. For rows $z, w \notin \{1, 2\}$ every trade of rows z, w along a $\rho_{z,w}$ -cycle or $\rho_{z,w}$ -arc changes rows z, w only, hence is legal in X . In particular:

Lemma 5.1 (exact fairness of column-pair status for generic rows). *Let $x, y \notin \{1, 2\}$ and let (q, c) be a column pair lying in two different cycles of $\rho_{x,y}$. The CGW switch at (q, c) for the rows x, y (trade of rows x, y along the one of the two cycles that contains the smaller minimum symbol) is a legal involution of X that toggles whether x, y lie in one column cycle of (q, c) , and preserves the cycle partition of $\rho_{x,y}$ and the separation status of every column pair with respect to $\{p, p'\}$. Hence, conditionally on everything determined by the other rows and by the cycle partition of $\rho_{x,y}$,*

$$\mathbb{P}_X[x, y \text{ in different column cycles of } (q, c)] = \frac{1}{2}.$$

Proof. As in Lemma 2.3 with $(1, 2)$ replaced by (x, y) : the trade swaps the entries of rows x, y on the columns of the chosen cycle $Q \ni q$, $Q \not\ni c$, so $\delta_{q,c} \mapsto \delta_{q,c} \circ (xy)$, which splits or joins the column cycles through x, y . It inverts Q in place, so the symbol sets of the two cycles are unchanged and the min-symbol rule re-selects Q : an involution. Rows 1, 2 are untouched, so type and mark are preserved. Inverting Q exchanges the two arcs between p and p' when $Q \ni p, p'$, so separation of any pair by $\{p, p'\}$ is preserved. (If Q contains p or p' the frame changes at rows x, y ; so frame-determined data are part of the conditioning only when Q avoids p, p' .) $\square$

A column turn, by contrast, changes the rows of a column cycle. Let (q, c) be a free column pair and Z one of its column cycles.

Lemma 5.2 (legal column turns). *The turn of Z (swapping the entries of columns q, c in the rows of Z) is a measure-preserving involution of X if and only if $|Z \cap \{1, 2\}| \in \{0, 2\}$; when $|Z \cap \{1, 2\}| = 2$ it conjugates σ by (qc) and keeps the type and the mark. It preserves the columns p, p' (hence the frame and every frame-determined family of row pairs) and the column-cycle partition of (q, c) and of every column pair disjoint from $\{q, c\}$ (in particular of every pair of a matching containing (q, c)). For a row pair (x, y) with $|Z \cap \{x, y\}| = 1$ it replaces $\rho_{x,y}$ by $\rho_{x,y} \circ (qc)$ (if $x \in Z$) or $(qc) \circ \rho_{x,y}$ (if $y \in Z$), which toggles the crossed/parallel status of (x, y) exactly when (q, c) is separated by $\{p, p'\}$ in $\rho_{x,y}$ (different arcs when crossed, different cycles $Q_p, Q_{p'}$ when parallel), and leaves (q, c) separated afterwards; with $|Z \cap \{x, y\}| \in \{0, 2\}$ the status is unchanged. Turns of cycles of column pairs from a fixed matching $\mathcal{M}$ of the free columns pairwise commute.*

Proof. A column cycle is a Latin trade (Lemma 2.2), so the turn is a bijection of Latin squares and an involution, and it preserves the cycle partition of (q, c) ; column pairs disjoint from $\{q, c\}$ are untouched, and so are columns p, p' . Rows 1, 2 change iff Z meets them; if $Z \ni 1, 2$ both rows swap their entries at q, c , so $\sigma' = (qc)\sigma(qc)$, whose type is that of σ and in which p, p' (fixed by (qc)) keep their distance. If Z meets exactly one of rows 1, 2, $\sigma' = \sigma \circ (qc)$ or $(qc) \circ \sigma$, which changes the type. The effect on $\rho_{x,y}$: if $x \in Z$ and $y \notin Z$ the entries of row x at q, c are exchanged, so $\rho'_{x,y} = \rho_{x,y} \circ (qc)$, and symmetrically; multiplying by a transposition of two points on one cycle splits it at those points, so p, p' are separated iff q, c lie on different arcs

between them, and multiplying by a transposition of two points on different cycles merges the cycles, so p, p' are joined iff q, c lie one in each of their cycles; in both cases q, c lie one on each side afterwards. Turns of cycles of disjoint column pairs act on disjoint sets of cells; two cycles of the same pair have disjoint row sets. $\square$

(c) The product bound. Fix a matching $\mathcal{M}$ of the free columns, independent of the square, and a frame-determined family $\mathcal{G}$ of row pairs (x, y) , $x, y \notin \{1, 2\}$, whose x -rows are pairwise distinct and disjoint from its y -rows (a block of the ladder, where the pairs are even disjoint; or a union of diagonals D_c with distinct x -rows, which may share y -rows). For $(x, y) \in \mathcal{G}$ call a column pair $(q, c) \in \mathcal{M}$ a *clean coordinate* of (x, y) if the column cycle Z of (q, c) through x has length at most ℓ_1 (a fixed constant), satisfies $|Z \cap \{1, 2\}| \in \{0, 2\}$, and contains no row of $\mathcal{G}$ other than x (no other x -row and no y -row, in particular not y). Let $\mathcal{T}_{x,y}$ be the set of clean coordinates of (x, y) and $\eta_{x,y} \in \{0, 1\}^{\mathcal{T}_{x,y}}$. By Lemma 5.2 the turns of all clean coordinates of all pairs of $\mathcal{G}$ commute, generate a group acting freely on X , preserve $\mathcal{G}$ and the sets $\mathcal{T}_{x,y}$, and

$$\rho_{x,y}(\eta) = \rho_{x,y}(0) \circ \prod_{(q,c) \in \mathcal{T}_{x,y}: \eta_{x,y}(q,c)=1} (q, c),$$

the bit $b_{x,y}(\eta) = [(x, y) \text{ parallel}]$ depending on $\eta_{x,y}$ only. Let the *orbit weight* of (x, y) be

$$\bar{y}_{x,y} = \mathbb{E}_{\eta_{x,y}} \left[\frac{\#\{(q, c) \in \mathcal{T}_{x,y} : (q, c) \text{ separated by } \{p, p'\} \text{ in } \rho_{x,y}(\eta)\}}{|\mathcal{T}_{x,y}|} \right]$$

(0 if $\mathcal{T}_{x,y} = \emptyset$), a function of the orbit, and $Y_{\mathcal{G}} = \sum_{(x,y) \in \mathcal{G}} \bar{y}_{x,y}$.

Proposition 5.3 (product bound). *Conditionally on the orbit, the bits $b_{x,y}$, $(x, y) \in \mathcal{G}$, are independent, and*

$$\mathbb{P}[(x, y) \text{ crossed} \mid \text{orbit}] \leq 1 - \frac{1}{2} \bar{y}_{x,y}, \quad \mathbb{P}[\text{no pair of } \mathcal{G} \text{ is flippable} \mid \text{orbit}] \leq \prod_{(x,y) \in \mathcal{G}} (1 - \frac{1}{2} \bar{y}_{x,y}) \leq e^{-Y_{\mathcal{G}}/2}.$$

Consequently $\mathbb{P}_X[\text{Flip}_{\mathcal{G}} = \emptyset] \leq \mathbb{E}_X[e^{-Y_{\mathcal{G}}/2}]$.

Proof. The action is free: distinct column pairs act on disjoint cells, and two clean coordinates of different pairs at the same (q, c) are the cycles through two distinct x -rows, each avoiding the other, hence different cycles; the sets $\mathcal{T}_{x,y}$ are orbit-invariant by Lemma 5.2 (row sets of the cycles, their meeting $\{1, 2\}$, and the frame-determined family are preserved). The uniform measure on X restricted to an orbit is therefore uniform on the orbit, which is the product of the cubes $\{0, 1\}^{\mathcal{T}_{x,y}}$. A clean coordinate of (x, y) contains x and not y , and a clean coordinate of another pair contains neither x nor y (it avoids all x -rows but its own and all y -rows), so $b_{x,y}$ depends on $\eta_{x,y}$ alone, whence the independence. Fix (x, y) and $Z = (q, c) \in \mathcal{T}_{x,y}$. Whether Z is separated in $\rho_{x,y}(\eta)$ does not depend on η_Z (Lemma 5.2: the turn of Z leaves Z separated), and when it is, the two values of η_Z give opposite bits. Hence $\mathbb{P}[b_{x,y} = 0] \leq 1 - \frac{1}{2} \mathbb{P}_{\eta}[Z \text{ separated}]$ for every Z , and averaging the right side over $Z \in \mathcal{T}_{x,y}$ gives the first bound; the second is the product of the first over the independent bits. $\square$

Hypothesis 5.4 (orbit genericity). *There are constants $\kappa_0 > 0$ and ℓ_1 such that, for the fixed matching $\mathcal{M}$ of Proposition 5.3 and uniformly in λ, α, β , with $\ell_0 = n/\log^2 n$: (i) for the sub-ladder I of the first $\min(K_0 - 1, \lceil n/8 \rceil)$ ladder pairs, $\mathbb{P}[K_0 \geq \ell_0, Y_I < \kappa_0 | I] = o(1/\log n)$; (ii)*

for a B -instance with $d_{12} = \ell \leq n/4$ and $|C_1| \geq n/2 + \ell$, with $S = \{2\ell t : 0 \leq t \leq (n/2 - \ell)/(2\ell)\}$ (offsets at gaps 2ℓ , so that the x -rows $\pi^{-(c+i)}(1)$, $c \in S$, $1 \leq i < \ell$, are pairwise distinct, $|S| \geq n/(4\ell)$, and the diagonals D_c , $c \in S$, involve at most $n/4 + \ell$ rows), and cleanliness taken with respect to all rows of $\bigcup_{c \in S} D_c$: the event that fewer than half of the D_c , $c \in S$, have $Y_{D_c} \geq \kappa_0(\ell - 1)$ has probability $o(1/\log n)$ (as a single event over all ℓ).

Proposition 5.5. *Hypothesis 5.4 implies (L), hence the weak CGW conjecture.*

Proof. Put $\ell_0 = n/\log^2 n$. By Theorem 4.2, $|\mathbb{P}[B] - \mathbb{P}[A]| \leq \mathbb{P}[\text{Flip} = \emptyset] \leq \mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] + \mathbb{P}[A, K_0 < \ell_0] + \mathbb{P}[B, K_0 < \ell_0]$. Long ladders: by Proposition 5.3, on $\{K_0 \geq \ell_0\}$ and off the exceptional event of (i), $\mathbb{P}[\text{Flip} = \emptyset \mid \text{orbit}] \leq e^{-\kappa_0|I|/2} \leq e^{-\kappa_0\ell_0/4} = o(1/\log n)$. Short arcs: fix ℓ and apply Proposition 5.3 to the family $\bigcup_{c \in S} D_c$: its x -rows $\pi^{-(c+i)}(1)$ lie in the disjoint intervals of depths $[c+1, c+\ell-1]$, $c \in S$, so they are pairwise distinct, and they lie on the frame arc from 2 to 1 while the y -rows $\pi^{-i}(2)$ lie on the arc from 1 to 2, so the two sets are disjoint. Given the orbit the bits of all pairs are independent, hence so are the diagonals' flippabilities, and off the exceptional event at least $|S|/2$ of the diagonals are flippable with conditional probability $\geq 1 - e^{-\kappa_0(\ell-1)/2} \geq 4\theta_0$, $\theta_0 := \frac{1}{4}(1 - e^{-\kappa_0/2})$. The number of flippable offsets in S then has conditional mean $\geq 2\theta_0|S|$, and by Chernoff the fraction θ_S is $\geq \theta_0$ except with conditional probability $\leq e^{-\theta_0|S|/4} \leq e^{-\theta_0 \log^2 n/16}$ (as $|S| \geq n/(4\ell_0) = \log^2 n/4$), summable over $\ell < \ell_0$ to $o(1/n)$. The identity behind (4) holds for the average over $c \in S$ as well as over all c , so $\sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta_S \geq \theta_0] \leq 36\ell_0/((n-2)\theta_0) = O(1/\log^2 n)$; the part $|C_1| < n/2 + \ell$ costs $\leq 8/n$ per ℓ by Proposition 4.7, and $\mathbb{P}[A, K_0 < \ell_0] \leq 72\ell_0/n$ by Proposition 4.6; the case $d_{21} = \ell$ is symmetric. Altogether $\mathbb{P}[\text{Flip} = \emptyset] = o(1/\log n)$. $\square$

What the hypothesis says. It replaces a lower-tail statement about rare short cycles (a ladder pair is flippable as soon as its $\rho_{x,y}$ -cycle through p is short and avoids p' , an event of probability $\approx \ell'/n$ per pair for cycles of length $\leq \ell'$) by a statement about a quantity that is typically of constant size: $\bar{y}_{x,y}$ is the average, over the cube of a few column turns that fix rows y and 1, 2 and the whole family, of the fraction of those turns' column pairs that are separated by $\{p, p'\}$ in the row pair (x, y) . It is a *genericity* statement — the short column cycles of the matching pairs through x are not aligned with the arcs of $\rho_{x,y}$ at p, p' — together with concentration of the sum over the family; concentration of the *flippabilities* themselves is no longer required, since Proposition 5.3 makes them conditionally independent. It still contains inputs of the same kind: a positive orbit weight needs the cycles of $\rho_{x,y}$ through p, p' to be macroscopic (the arc-trapping input (α) of §6), and the genericity itself. The pure permutation core is: for a permutation ρ_0 and a matching $\mathcal{T}$ of transpositions on points other than p, p' , with $\tau(\eta)$ the product of a uniformly random subset of $\mathcal{T}$, bound $\mathbb{P}_\eta[p, p' \text{ separated in } \rho_0\tau(\eta)]$ from below. For an adversarial $\mathcal{T}$ (e.g. all pairs at ρ_0 -distance 2, none straddling p or p') the probability is 0, so the content is that the clean coordinates of a random square are not adversarial; and the moves that re-randomise the clean coordinates without touching $\rho_{x,y}$ — the free row trades of Lemma 5.1 — have availabilities that are again same-cycle events of generic row pairs. This is the recursion: every availability is a cycle-avoidance event, and the exact identities in X (Lemma 5.1, Proposition 5.3, and the one-step switchings of the marked space) are all fair. What Proposition 5.3 adds is that the fairness can be *compounded* over a family: a product of conditionally independent coins, so that a constant per-pair genericity gives exponential decay in the family size.

(d) **Data.** All scripts and logs referred to below are in the repository [5], under `s13*.py` and `runs/s13/orbit/`. The orbit test (600, 600 and 360 ladder pairs at $n = 30, 50, 100$, marks drawn as in X , $\ell_1 \leq 4, 5, 8$; here the matching is chosen greedily among all clean short pairs of the instance, which over-counts the coordinates of a fixed matching by a factor $\approx n/\ell_1$ — observed 10.8 and 14.3 — but tests the same genericity):

- the length of the column cycle of (q, c) through x is spread like a uniform variable on $\{2, \dots, n\}$ (about $1/n$ per length); the turn is legal for 98% of the 2-cycles and 64% of the long cycles, clean for 97% resp. 27% ($n = 100$);
- genericity: $\mathbb{P}[(q, c) \text{ separated} \mid \text{clean, length } \ell] = 0.199, 0.198, 0.193, 0.201, 0.201, 0.198, 0.197$ for $\ell = 2, \dots, 8$, against $\mathbb{P}[\text{separated}] = 0.195$ over all column pairs ($n = 100$; at $n = 50$: 0.186–0.194 against 0.184); the dependence on the arc lengths of $\rho_{x,y}$ at p, p' was not binned;
- the orbit coin: with $|\mathcal{T}| \approx 42$ coordinates ($n = 100$) the orbit average of the parallel bit is 0.478 starting from a crossed square and 0.523 from a parallel one (at $n = 30$: 0.398/0.594 with $|\mathcal{T}| \approx 10$; at $n = 50$: 0.447/0.561 with $|\mathcal{T}| \approx 18$); at $n = 100$, 243 of the 360 per-instance orbit averages lie in $[0.4, 0.6)$.

So $\bar{y} \approx 0.2$ with no detectable dependence on the cycle length, and the orbit coin is fair to within the mixing allowed by $\approx 0.2|\mathcal{T}|$ separated coordinates. With the *fixed* matching of consecutive free columns, as in Proposition 5.3: $|\mathcal{T}| = 1.7$ resp. 3.0 clean coordinates per pair on average ($\approx 0.9(\ell_1 - 1)/2$, as the uniform spread of the lengths predicts), $\mathcal{T} = \emptyset$ for 15% resp. 4% of the pairs, and $\mathbb{E}[\bar{y}] = 0.159$ resp. 0.189. So the per-pair factor of Proposition 5.3 is $1 - \frac{1}{2}\bar{y} \approx 0.9$ on average, and Hypothesis 5.4(i) is the statement that the average over the sub-ladder does not collapse; the data put the average of $\bar{y}$ at 0.13–0.19, so the hypothesis asks for some κ_0 well below that, and the lower tail is examined in (d').

(d') **The concentration input (γ): data.** The covariance test (543 and 286 instances at $n = 50, 100$) computes, with the fixed matching and cleanliness with respect to all rows of the sub-ladder I of the first $\min(K_0 - 1, n/8)$ pairs, every $\bar{y}_k$, $k \in I$, and $Y_I = \sum_{k \in I} \bar{y}_k$: mean $|I| = 5.2$ resp. 9.9, $\mathbb{E}[\bar{y}] = 0.130$ resp. 0.135, $\text{Var}(\bar{y}) = 0.084$ resp. 0.080, and $\text{Var}(Y_I) = 0.43$ resp. 0.90 against $|I|\text{Var}(\bar{y}) = 0.44$ resp. 0.79: the ratio 0.99 resp. 1.14 (both including the variation of $|I|$, which at $n = 100$ contributes $\approx \mathbb{E}[\bar{y}]^2 \text{Var}(|I|) \approx 0.2$) shows no detectable positive correlation between the orbit weights of different ladder pairs. So (γ) is, numerically, an independence statement. The orbit weight has an atom of ≈ 0.8 at 0 (few coordinates per pair with a fixed matching), and the lower tail $\mathbb{P}[Y_I < \kappa_0 | I|]$ at these sizes (0.18 at $\kappa_0 = 0.05$, $n = 100$) is dominated by the instances with $|I| \leq 4$.

6 What remains

Hypothesis 5.4 has three inputs: (α) the cycles of ρ_{x_k, y_k} through p, p' are macroscopic for a constant fraction of the ladder pairs (both arcs when they coincide, both cycles when they do not); (β) *genericity*: the clean coordinates of a ladder pair are separated by $\{p, p'\}$ with probability bounded below; (γ) concentration of $\sum_k \bar{y}_k$ over the family. Numerically (γ) is an independence statement (the orbit weights of different ladder pairs are uncorrelated to within sampling error), (β) holds with the same constant 0.19–0.20 at every cycle length, and (α) holds

with the lengths spread uniformly. All three are instances of one statement:

$$(U) \quad c \leq \mathbb{P}[\text{two given pairs of lines are interleaved in the cycle structure of a third pair}] \leq 1-c$$

in X — where two column pairs are interleaved in a row pair (x, y) if they alternate on a common cycle of $\rho_{x,y}$ or lie crosswise in two of its cycles, and dually for row pairs in a column pair. The pure permutation part of the problem is settled:

(e) The permutation problem (II), and the one statement everything reduces to.

Fix a permutation ρ_0 of the columns, two columns $p \neq p'$, and a set $\mathcal{T}$ of pairwise disjoint transpositions avoiding p, p' ; for $\eta \in \{0, 1\}^{\mathcal{T}}$ let $\tau(\eta)$ be the product of the transpositions with $\eta_Z = 1$. Call $Z = (q, c)$ *crossing* in a permutation ρ (“separated” in Lemma 5.2) if $\{q, c\}$ and $\{p, p'\}$ are interleaved in the cycle structure of ρ : q, c on different arcs of a common cycle through p, p' , or q in the cycle of p and c in the (different) cycle of p' , or vice versa. By Lemma 5.2, $\rho \mapsto \rho \circ Z$ toggles the separation of p, p' exactly when Z is crossing in ρ , and Z is crossing in $\rho \circ Z$ iff it is crossing in ρ . Hence, for η uniform and $b(\eta) = [p, p' \text{ separated in } \rho_0 \tau(\eta)]$,

$$\mathbb{P}_\eta[b = 1] \geq \frac{1}{2} \max_{Z \in \mathcal{T}} \mathbb{P}_\eta[Z \text{ crossing in } \rho_0 \tau(\eta)] \geq \frac{1}{2} \mathbb{E}_\eta \left[\frac{\#\{Z \text{ crossing in } \rho_0 \tau(\eta)\}}{|\mathcal{T}|} \right], \quad (6)$$

and $b \equiv 0$ on the cube iff no Z is crossing in any $\rho_0 \tau(\eta)$. Let ρ_0 be crossed, with arcs P_1 (from p to p') and P_2 , and form the graph whose vertices are P_1, P_2 and the other cycles of ρ_0 and whose edges are the transpositions of $\mathcal{T}$ joining different vertices. Then $b \equiv 0$ iff P_1 and P_2 lie in different components. If they are connected by a path $P_1 = V_0, V_1, \dots, V_m = P_2$ with edges $Z_1, \dots, Z_m$, apply $Z_1, \dots, Z_{m-1}$ in order: Z_i , with one end on the arc from p to p' and the other in V_i , inserts V_i into that arc ($\rho \circ (qc)$ sends $q \mapsto \rho(c)$, runs around V_i , and returns to $\rho(q)$), P_2 is untouched, and Z_m then joins the arc to P_2 and is crossing. If they are disconnected, let K_1 be the union of the vertices in the component of P_1 and K_2 that of P_2 : every transposition of $\mathcal{T}$ has both ends in K_1 , both in K_2 or both outside, so $\tau(\eta)$ preserves K_1 , while $\rho_0(K_1) = (K_1 \setminus \{p\}) \cup \{p'\}$; the $\rho_0 \tau(\eta)$ -orbit of p therefore stays in K_1 until it reaches p' , and it cannot close at p before that, since $\rho_0 \tau(x) = p$ forces $\tau(x) = \rho_0^{-1}(p) \in P_2 \subseteq K_2$. In particular a single crossing Z at $\eta = 0$ gives $\mathbb{P}_\eta[b = 1] \geq \frac{1}{2}$ on the orbit of Z alone, and a family of chords at ρ_0 -distance 2, none straddling p or p' , is a disconnected example. One may equally use one coordinate per pair: with Z_k the first clean coordinate of the k -th pair in a fixed order of the matching (orbit-invariant, since cleanliness of every matching pair is), the proof of Proposition 5.3 applied to the sub-family $\{Z_k\}$ gives

$$\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X \left[\prod_{k \in I} \left(1 - \frac{1}{2} X_k\right) \right], \quad X_k = [Z_k \text{ crossing in } \rho_{x_k, y_k}], \quad (7)$$

and (β) becomes: $\mathbb{P}_X[X_k = 1] \geq c$ for most k , i.e. the column pair selected by the column-cycle structure through x_k is interleaved with $\{p, p'\}$ in the row-cycle structure of (x_k, y_k) with probability bounded below.

Three facts locate the difficulty. (i) Without conditioning on the type of rows 1, 2, (U) for four generic columns in a generic row pair is elementary in its upper half: by conjugation invariance the four columns are uniform points given the cycle type of $\rho_{x,y}$, and the interleaving probability is $\frac{1}{3} \sum_i (m_i)_4 / (n)_4 + 2 \sum_{i \neq j} (m_i)_2 (m_j)_2 / (n)_4$ over the cycle lengths m_i , which is at most $\frac{1}{3} + O(1/n)$ for every type (with equality for the n -cycle) and, by Jensen, at least

$\frac{1}{3}\mathbb{P}[W \mid \text{type}]^3 - O(1/n)$, where W is the event that two given columns share a cycle of $\rho_{x,y}$; the lower half therefore holds exactly when $\rho_{x,y}$ has macroscopic cycles with probability bounded below (it fails for the type $(2^{n/2})$, where the formula is $O(n^{-2})$), which is the content of (α) for a generic row pair. (ii) For generic rows the dual statement is trivial: the number of row pairs interleaved with $\{1, 2\}$ in a column pair is at most $(n-2)^2/4$, half of all pairs, deterministically, and rows other than 1, 2 are exchangeable in X . What is needed is the same for a row pair selected by its own row-cycle structure (the ladder pair, at the columns where its cycle through p meets a joining pair), and exchangeability is lost exactly there. (iii) Inside X the exact tools are the free trades of rows other than 1, 2 (which preserve interleaving) and the column-cycle turns (which are legal exactly off the interleaved configurations), and every identity they produce is fair: it equates two expectations with random weights whose lower bounds are again statements of the form (U). Lower bounds on “different cycles” events are, in the language of [1], of the type of the direction of their Lemma 3.12 whose proof has the gap: they require a canonical bounded-multiplicity repair of the same-cycle configurations, and the repairs available in X (trades or cross-switches of a free pair of third rows) have unbounded multiplicity. What the orbit method adds to this is that the fairness compounds: a constant per-pair genericity gives exponential decay in the family size, which is why a hypothesis of constant strength suffices for a conclusion of strength $o(1/\log n)$.

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